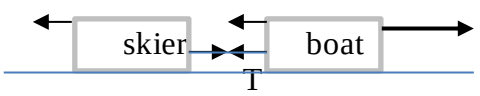
Question			Answer	Marks	AO	Guidance
10	(a)		400N 1200N 2400N  	B1 B1 [2]	1.1 1.1	Driving force and common tension correct and labelled (boat shown on right – allow mirror image) Allow T marked as a thrust if consistent with equations in 10bi and 10bii Both resistances correct and labelled
10	(b)	(i)	N2L for skier $T-400=65a$	B1 [1]	1.1	Allow $-T-400=65a$ if consistent with their diagram
10	(b)	(ii)	N2L for the boat $2400-1200-T=985a$ $1200-T=985a$	M1 A1 [2]	1.1 1.1	Attempt to use N2L with $m=985$ for the boat. Allow missing-or incorrect resistance. (do not allow for weight included) Allow 400 for resistance if 1200 used in (i) Allow -T used for T if consistent with their diagram. Allow if not simplified Allow SC1 for $2400-1200-400=1050a$
10	(c)		Add equations to give acceleration $\frac{800}{1050}=0.762\,\text{m s}^{-2}$	B1 [1]	1.1	Cao Allow if resistances interchanged
10	(d)		$T_{max}=65g$ $65g-400=65a$ gives $a=3.65\ \text{m s}^{-2}$	M1 A1 [2]	3.1b 1.1	Substitute 65g for T in their equation from (b)(i) Accept awrt 3.6 or 3.7